



Code ▾

Outliers, Leverage, and Influential Observations

AUTHORS

Julián David Rojas Aguilar
Ignacio Sarmiento-Barbieri

AFFILIATION

Universidad de los Andes
Universidad de los Andes

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Regression Diagnostics and Sensitivity Analysis

Regression models can reveal data errors, unusual covariate combinations, or aspects of the *data-generating process* (DGP) that the specification does not capture¹. This notebook distinguishes three concepts that are related but are not interchangeable:

Concept	Diagnostic question	Measure
Outlier	Is the outcome unusual after conditioning on the regressors?	Residuals, $\hat{\epsilon}_i = y_i - \hat{y}_i$
High-leverage observation	Is the regressor vector unusual relative to the rest of the sample?	Diagonal of the projection matrix, h_{ii}
Influential observation	Does omitting the observation change the fitted model?	Function of $\hat{\epsilon}_i$ and h_{ii}

A high-leverage observation does not have to be an outlier, and an outlier does not have to be a high-leverage observation. Influence arises when an observation combines an unusual regressor vector with a sufficiently large residual. The objective is not to delete observations mechanically, but to understand why they matter and to assess how sensitive the results are to them.

Acknowledgement: Most of the materials used in this complementary session were contributed by previous TAs, especially [Eduard Martínez](#) , [Lucas Gómez](#) , Andrés Rengifo, Valentina Laverde, and Gustavo Castillo. We are truly grateful for their help.

Empirical Example: Hours Worked

Suppose we want to study labor supply at the intensive margin: among people who work, how do observed characteristics relate to their total hours worked? We are particularly interested in gender differences and whether the association between household composition and hours differs between men and women.

We begin with the general model:

$$\text{Hours}_{if} = \alpha + X_i\beta + D_i\delta + F_f\phi + \varepsilon_{if},$$

where Hours_{if} denotes the hours worked by individual i in family f . The vector X_i contains variables related to the potential gains from working more, D_i contains individual demographic characteristics, and F_f contains family characteristics. We use this model as an empirical setting for studying diagnostics.

Initial Configuration

We use the `pacman` package to simplify package management.

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```
1 # Install pacman if necessary, then load the packages used in this session.
2 if (!require(pacman)) install.packages("pacman")
3 library(pacman)
4
5 p_load(
6   rio,          # Import data from a URL.
7   tidyverse,    # Manipulate data and create figures.
8   MASS,         # Calculate residuals.
9   conflicted    # Resolve conflicts between functions with the same name.
10 )
11
12 conflict_prefer(name = "select", winner = "dplyr")
13 conflict_prefer(name = "filter", winner = "dplyr")
14
15 options(scipen = 999)
```

Importing Data

We are going to import a portion of the GEIH data that will be used in the first problem set.

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```
1 db_geih <- rio::import("https://raw.githubusercontent.com/ignaciomsarmiento/datasets/main/GE
2   as_tibble()
```

The sample contains people interviewed in Bogotá. The combination of building, household, and person-order identifiers distinguishes individuals. The [data dictionary](#) documents the survey variables.

Cleaning the Dataset

Our unit of analysis is an employed adult with positive reported hours. Before imposing that individual-level restriction, we construct household variables using all observed household members.

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```
1  # The names of this vector are the standardized names used below; its values
2  # are the corresponding names in the original data.
3  text_vars <- c('id_building' = 'directorio', 'id_household' = 'secuencia_p',
4                'id_person' = 'orden', 'num_age' = 'age', 'bin_male' = 'sex',
5                'bin_employed' = 'ocu', 'bin_formal' = 'formal',
6                'num_hoursWorked' = 'totalHoursWorked', 'cat_job' = 'oficio',
7                'cat_relationship' = 'p6050', 'num_labIncome' = 'y_total_m')
8
9  db_geih <- db_geih |>
10   select(all_of(text_vars))
```

Potential Gains from Working More

We use mean labor income within occupation-gender cells as an illustrative proxy for the potential gains from working more. We express the proxy in millions of Colombian pesos (COP). The occupation code has 99 possible values, so we retain the codes rather than create labels, which are not central to this exercise.

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```
1  db_occupAndGender <- db_geih |>
2    group_by(cat_job, bin_male) |>
3    summarise(num_potentialGain = mean(num_labIncome, na.rm = TRUE),
4              .groups = "drop") |>
5    mutate(num_potentialGain = num_potentialGain/1000000)
6
7  db_occupAndGender |>
```

```
8 select(cat_job, bin_male, num_potentialGain) |>
9 head(n = 4)
```

cat_job	bin_male	num_potentialGain
<int>	<int>	<dbl>
2	0	3.9079942
2	1	3.7369682
3	0	0.9528268
3	1	1.5083546

4 rows

We merge the cell-level proxy into the individual-level data.

▼ Show/hide code

```
1 db_geih <- db_geih |>
2 inner_join(db_occupAndGender, by = c("cat_job", "bin_male"))
```

Number of Children per Household

We count household members younger than 18 before restricting the sample to employed adults².

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```
1 db_geih <- db_geih |>
2 mutate(bin_minor = ifelse(test = num_age < 18, yes = 1, no = 0)) |>
3 group_by(id_building, id_household) |>
4 mutate(num_minors = sum(bin_minor, na.rm = TRUE)) |>
5 ungroup() |>
6 select(-bin_minor)
7
8 db_geih |>
9 arrange(desc(id_building)) |>
10 select(id_building, id_household, num_age, num_minors) |>
11 tail(n = 6)
```

id_building	id_household	num_age	num_minors
<int>	<int>	<int>	<dbl>
4514332	1	4	2
4514332	1	7	2
4514332	1	32	2
4514332	1	35	2
4514331	1	29	0
4514331	1	36	0

6 rows

Household Head and Sample Restriction

We identify the household head and then retain employed respondents aged 18 or older with positive reported hours. For this exercise, we use complete cases for the variables in the regression. You should, however, document the causes and patterns of missingness rather than drop observations by default.

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```
1 db_geih <- db_geih |>
2   mutate(bin_head = ifelse(test = cat_relationship == 1, yes = 1, no = 0)) |>
3   filter(bin_employed == 1, num_age >= 18, num_hoursWorked > 0) |>
4   drop_na()
```

Regression Model

The empirical specification is

$$\text{Hours}_{if} = \alpha + \beta_1 \text{PotentialGain}_i + \beta_2 \text{Minors}_f + \beta_3 \text{Male}_i + \beta_4 (\text{Minors}_f \times \text{Male}_i) + \beta_5 \text{Head}_i + \beta_6 \text{Age}_i + \varepsilon_{if}$$

▼ Show/hide code

```
1 model_formula <- num_hoursWorked ~ num_potentialGain + num_minors * bin_male +
2   bin_head + num_age
3
4 linear_model <- lm(formula = model_formula, data = db_geih)
5
6 summary(linear_model)
```

Call:

```
lm(formula = model_formula, data = db_geih)
```

Residuals:

Min	1Q	Median	3Q	Max
-48.243	-5.163	-0.048	4.483	77.854

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

```

(Intercept)      48.04176    1.52653   31.471 < 0.0000000000000002 ***
num_potentialGain -0.37346    0.35355   -1.056      0.291014
num_minors       -0.07854    0.58393   -0.134      0.893028
bin_male         2.79051    1.03840    2.687      0.007293 **
bin_head         2.86642    0.82740    3.464      0.000548 ***
num_age          -0.09064    0.03031   -2.990      0.002841 **
num_minors:bin_male 1.42508    0.79219    1.799      0.072257 .

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.77 on 1335 degrees of freedom

Multiple R-squared: 0.04684, Adjusted R-squared: 0.04256

F-statistic: 10.93 on 6 and 1335 DF, p-value: 0.000000000006454

In this application, the control set contains the number of minors, gender, their interaction, household-head status, and age. Recall that the R expression `num_minors * bin_male` includes `num_minors`, `bin_male`, and their interaction. If you only want to add the interaction between two variables, you should use `num_minors:bin_male`.

Geometric Interpretation of OLS

The geometry of OLS clarifies why some observations have more potential to affect the fitted model. Let $\mathbf{y} \in \mathbb{R}^n$ and let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a full-rank design matrix that includes the intercept. The columns of $\mathbf{X}$ span:

$$\text{Col}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}$$

OLS selects the vector in this column space that is closest to $\mathbf{y}$. The fitted values and residuals are, respectively,

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}_\mathbf{X}\mathbf{y}$$

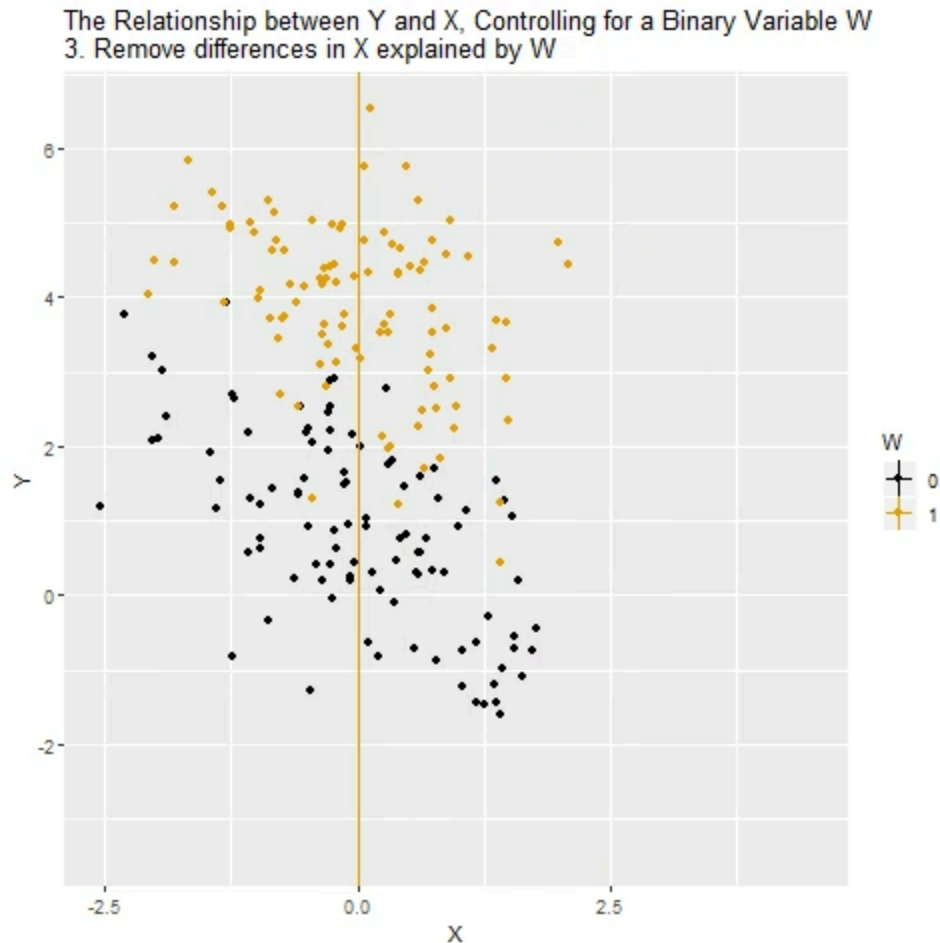
$$\hat{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}} = [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y} = \mathbf{M}_\mathbf{X}\mathbf{y}$$

Intuitively, $\mathbf{P}_\mathbf{X}$ shows what $\mathbf{X}$ can explain, and $\mathbf{M}_\mathbf{X}$ removes what $\mathbf{X}$ can explain. The Frisch–Waugh–Lovell theorem provides a useful application. Suppose that:

$$\mathbf{y} = \beta\mathbf{x} + \gamma\mathbf{z} + \varepsilon$$

To recover the OLS coefficient on $\mathbf{x}$, we may:

1. Remove the variation in $\mathbf{x}$ explained by $\mathbf{z}$ using $\tilde{\mathbf{x}} = \mathbf{M}_z \mathbf{x}$.
2. Remove the variation in $\mathbf{y}$ explained by $\mathbf{z}$ using $\tilde{\mathbf{y}} = \mathbf{M}_z \mathbf{y}$.
3. Regress $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{x}}$ to recover $\hat{\beta}$.



Intuition behind the Frisch--Waugh--Lovell theorem.

Source: Taken from Nick Huntington-Klein, in <https://nickchk.com/causalgraphs.html>

Pause animation

Example with Our Regression Model

Suppose that our target is β , the coefficient on the potential-gains proxy in the regression model above. We can estimate β directly by including `num_potentialGain` and all the controls in a single regression. The Frisch–Waugh–Lovell (FWL) theorem shows that we can recover exactly the same coefficient by isolating the variation in potential gains and hours worked that the controls do not explain. The following code implements FWL in three steps:

1. Regress hours worked on the controls. The residuals, `y_tilde`, represent the part of hours worked that is not (linearly) explained by those controls.

2. Regress the potential-gains proxy on the same controls. The residuals, `x_tilde`, represent the part of potential gains that is not (linearly) explained by those controls.
3. Regress `y_tilde` on `x_tilde`. This regression uses only the variation that remains after removing the linear contribution of the controls from both variables.

▼ Show/hide code

```
1 y_tilde <- resid(lm(num_hoursWorked ~ num_minors * bin_male + bin_head + num_age,
2                     data = db_geih))
3 x_tilde <- resid(lm(num_potentialGain ~ num_minors * bin_male + bin_head + num_age,
4                     data = db_geih))
5
6 # Since we also residualized with respect to the intercept, we must tell R
7 # not to include another intercept.
8 fwl_model <- lm(y_tilde ~ 0 + x_tilde)
```

Because both residualization regressions include an intercept, `y_tilde` and `x_tilde` have sample means of zero. The expression `0 + x_tilde` therefore prevents R from adding a new intercept in the final regression.

We can now compare the coefficient on `num_potentialGain` from the full model with the coefficient on `x_tilde` from the residual-on-residual regression:

▼ Show/hide code

```
1 coef(linear_model)["num_potentialGain"]
```

```
num_potentialGain
-0.3734613
```

▼ Show/hide code

```
1 coef(fwl_model)["x_tilde"]
```

```
x_tilde
-0.3734613
```

The two values are numerically identical! Therefore, FWL changes how we compute the coefficient, not how we interpret it.

Regression Diagnostics

Leverage: Unusual Regressor Values

Let $\mathbf{x}_i'$ denote the i th row of the design matrix. The leverage of observation i is the i th diagonal element of the projection matrix:

$$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$$

Let's examine this value. Remember that the length of a vector can be measured by its norm:

$\|\mathbf{x}_i\|_2 = \sqrt{\mathbf{x}_i'\mathbf{x}_i}$ ³. However, the leverage formula also includes the term $(\mathbf{X}'\mathbf{X})^{-1}$.

Intuitively, this factor accounts for the scale and correlation structure of $\mathbf{X}$: directions with large variance are compressed, directions with small variance are expanded, and correlated directions are decorrelated. Under this transformation, the scatterplot of the observations becomes spherical.

Therefore, a high h_{ii} indicates that $\mathbf{x}_i$ lies far from the other observations. Formally, it measures the squared Euclidean distance from the origin after the predictors have been orthonormalized⁴.

Let's calculate the leverage of each observation and the mean leverage:

▼ Show/hide code

```
1 db_geih <- db_geih |>
2   mutate(num_leverage = hatvalues(linear_model))
3
4 num_meanLeverage <- mean(db_geih$num_leverage)
```

A rule of thumb defines high-leverage observations as those with leverage greater than two or three times the mean leverage. Let's remove these observations and estimate the regression model again.

▼ Show/hide code

```
1 num_cutoff <- 3*num_meanLeverage
2 num_cutoff
```

```
[1] 0.01564829
```

▼ Show/hide code

```
1 db_geih_filtered <- db_geih |>
2   filter(num_leverage <= num_cutoff)
3
4 linear_model2 <- lm(formula = model_formula, data = db_geih_filtered)
5
6 summary(linear_model2)
```

Call:

```
lm(formula = model_formula, data = db_geih_filtered)
```

Residuals:

Min	1Q	Median	3Q	Max
-47.942	-5.228	-0.028	4.430	77.759

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	47.84948	1.64786	29.037	< 0.0000000000000002 ***
num_potentialGain	-0.51178	0.44871	-1.141	0.25427
num_minors	0.30145	0.65398	0.461	0.64492
bin_male	2.94605	1.07530	2.740	0.00623 **
bin_head	2.78215	0.84547	3.291	0.00103 **
num_age	-0.08629	0.03094	-2.789	0.00536 **
num_minors:bin_male	1.14274	0.87646	1.304	0.19253

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.84 on 1295 degrees of freedom

Multiple R-squared: 0.04517, Adjusted R-squared: 0.04075

F-statistic: 10.21 on 6 and 1295 DF, p-value: 0.0000000004596

Finally, note that leverage depends only on the regressors, not on the outcome. Therefore, an observation with an unusual combination of regressor values has high leverage and the *potential* to affect the estimated coefficients. Let's examine the other determinant.

Outliers: Unusual Outcomes Conditional on Regressors

In a regression model, an outlier is an observation with an outcome far from its fitted value, $\hat{\epsilon}_i = y_i - \hat{y}_i$ —that is, after conditioning on the regressors. This may suggest that the observation was generated by a process different from the one we are interested in. For example, consider a data-entry mistake in which a person's salary is recorded as 10 million instead of 1 million. This produces a salary observation that does not reflect equilibrium in the labor market.

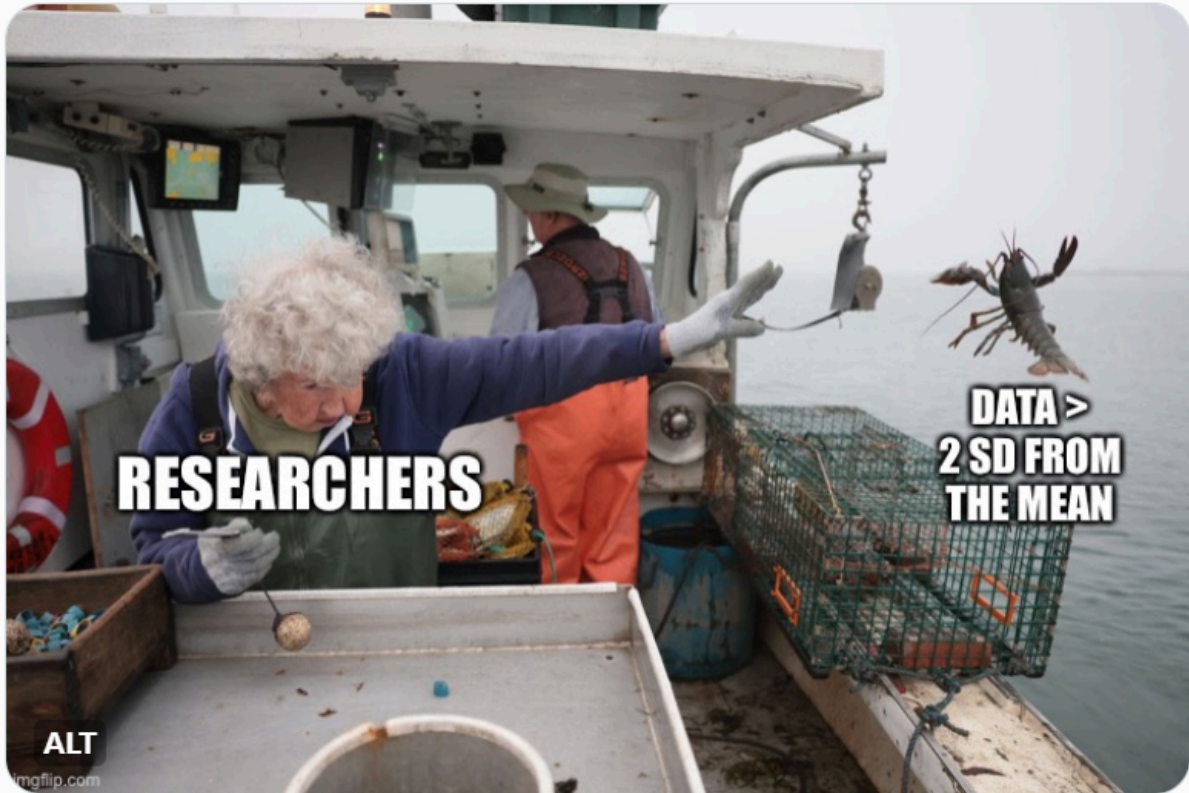
If we are interested in understanding how salaries are determined, we would like to drop observations that do not carry information about the labor market. This means that we would like to train our model using data from the same DGP. Therefore, if some data come from a different DGP, it might be useful to delete those observations, **but we must present evidence first!**



Chelsea Parlett-Pelleriti @ChelseaParlett · 16 sept. 2021

...

REPEAT AFTER ME: We get rid of outliers when their extremeness indicates they are not part of our population, we do not get rid of them solely because they are extreme.



46

596

4 mil

|||

🔖 ↗

Detecting Outliers in a Regression Model

An outlier may be detected by analyzing percentiles or the standard deviation of the residuals. However, a large residual does not by itself prove that the observation is erroneous or comes from another data-generating process. It may reflect a data-entry error, an omitted variable, nonlinearity, or a valid but rare case. The appropriate next step is to characterize the observation and investigate why it is unusual.

Let's do the exercise with the data from the GEIH. First, note that raw residuals are not directly comparable because their variances depend on leverage. We therefore use studentized residuals, which scale each residual using a leave-one-out estimate of its standard deviation.

▼ Show/hide code

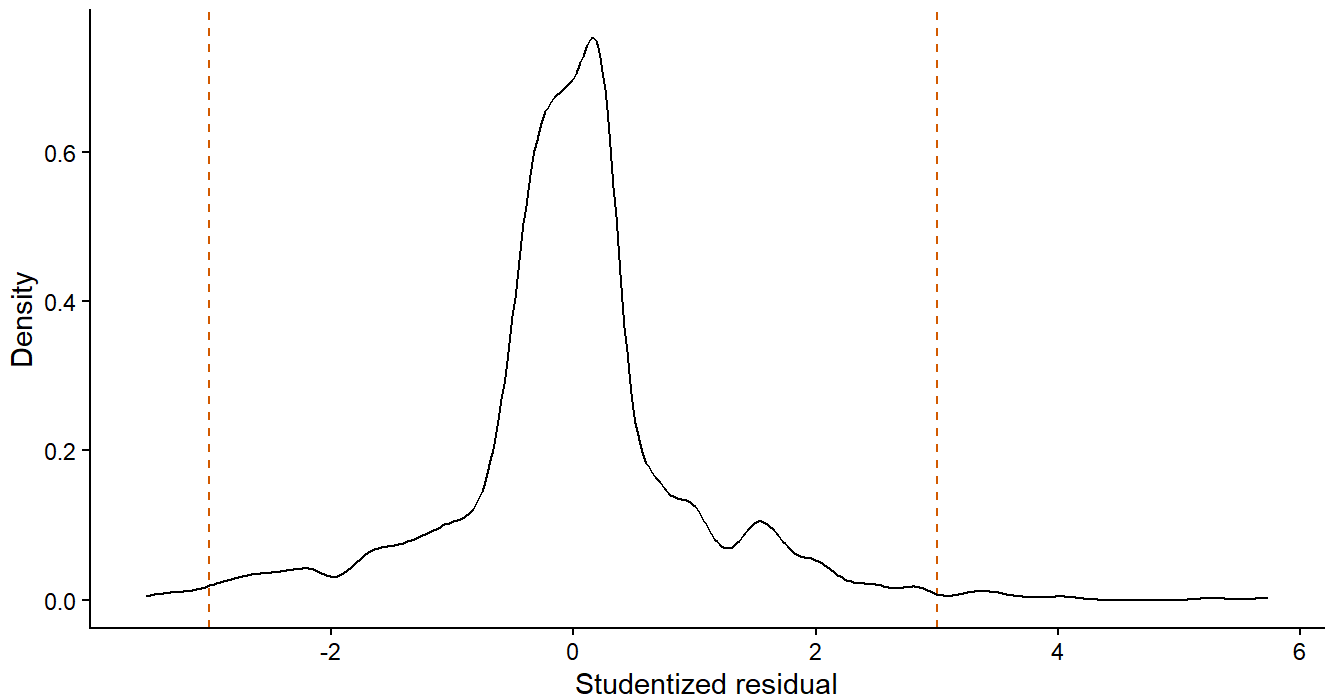
```
1 ggplot(data = db_geih |> mutate(num_studresid = studres(linear_model)),
2       mapping = aes(x = num_studresid)) +
3   geom_vline(xintercept = c(-3, 3), linetype = "dashed", color = "#D55E00") +
4   geom_density() +
5   labs(x = "Studentized residual",
```

```

6     y = "Density",
7     caption = "Note. Dashed lines mark the illustrative values -3 and 3." +
8     theme_classic()

```

Distribution of studentized residuals



Note. Dashed lines mark the illustrative values -3 and 3.

There are no strict statistical rules, only rules of thumb. For instance, we can study percentiles (1 and 99) or standard-deviation thresholds (below -3 or -2, or above 2 or 3). The objective, however, is to characterize these observations. Why are these outliers different from the other observations? What makes them special? Is it plausible for them to take these values? If so, we may be omitting a variable or misspecifying the model. If not, we may have found a data-entry mistake or an observation that does not belong to our DGP of interest—and may therefore consider deleting it.

Influence and Sensitivity Analysis

High leverage and large residuals together allow us to characterize influential observations. An influential observation has an important effect on the regression coefficients. When we exclude the i -th observation from the model, the regression coefficients change substantially.

Since the object of interest is the change when observation i is omitted, $\hat{\beta} - \hat{\beta}_{(-i)}$, we can show that:

$$\hat{\beta} - \hat{\beta}_{(-i)} = \left(\frac{\hat{\varepsilon}_i}{1 - h_{ii}} \right) (X'X)^{-1} x_i$$

This expression makes the distinction clear: high leverage magnifies the effect of a large residual, but a high-leverage observation with a residual near zero has little influence. This is why high-leverage observations have the *potential* to affect the estimated coefficients. If, for some i , the value $\hat{\varepsilon}_i = y_i - \hat{y}_i$ is sufficiently small, the observation will not be influential. In the extreme case in which $\hat{\varepsilon}_i = 0$, including or excluding the observation does not change the OLS estimates⁵.

Conclusion

We should study observations that satisfy both conditions: high leverage and a large residual. Begin by comparing the estimated coefficients when the group is included and excluded. An important change is evidence of sensitivity, not evidence that the restricted estimate is correct. Then, answer the following questions:

1. Who are the high-leverage observations? Are their values plausible?
2. Who are the outliers? Are their values plausible? Does the specification omit a relevant interaction or nonlinear term?

In summary, poor statistical reasoning would be to delete observations simply because their inclusion changes the estimated coefficients. A better practice is to provide a sensitivity analysis and investigate the observations that drive those changes. Outliers, high-leverage points, and influential observations are invitations to understand the data and the model—not mechanical instructions to remove observations.

Footnotes

1. If you have comments or suggestions, please send an email. They are welcome and will be considered in your final grade. [↩](#)
2. The association between children and hours worked may vary with the child's age. A richer specification could distinguish, for example, children aged 0–5 from those aged 6–17. [↩](#)
3. Another way to see the norm is the distance from the point x_i to the origin. [↩](#)
4. This means that every column has length one and that the columns are orthogonal to one another. [↩](#)
5. Note that influence is a property of an observation *relative to a particular model*. An observation may be influential under one specification and negligible under another. [↩](#)